

SEO Audit Toolkit & Website Analytics Platform

Proposed engineering roadmap — stack migration and feature build order

I'd like to propose the roadmap below for evolving the SEO Audit Toolkit into a full platform. Our current gap analysis surfaced 69 potential feature gaps against established competitors. Rather than tackle these in an arbitrary order, I've mapped each one against our planned infrastructure migration, so that feature work rides along with the infra we're already committed to, instead of getting bolted on afterward.

Two categories — Keyword/Backlink Intelligence and Content & AI Writing — don't depend on any of our planned infrastructure. They require entirely new data sourcing (e.g. a backlink index isn't something we can crawl into existence) and their own scoping conversation. I'm proposing we track these as a parallel backlog rather than force them into a specific version.

Gap Analysis: How Categories Map to Our Stack

Gap Category	Gaps	Depends On
Crawling & Technical SEO	13	Nothing new — extends existing crawler/audit code
Performance & Core Web Vitals	6	Nothing new — Lighthouse/PSI API + Postgres storage
AI Visibility / GEO (remaining)	2	Nothing new — external API polling + LLM call
Behavior & UX Analytics	13	New ingestion + ClickHouse + rrweb pipeline
Traffic & Conversion Analytics	19	Same pipeline, deeper
Platform & Integration	7	Production-facing; needs a stable public API first
Keyword/Backlink Intelligence	9	Nothing in our stack — third-party data, own vertical
Content & AI Writing	8	Nothing in our stack — LLM API calls, own vertical

The last two rows are the important call: they aren't blocked by NestJS, Postgres, or ClickHouse — they're entirely new product surfaces. Cramming them into the infra migration plan would make both worse.

v1 — Current (Done)

Baseline SEO Audit Toolkit is live and stable. This is our starting point for the roadmap below.

v2 — Backend Framework Migration

Express → NestJS + TypeScript, Puppeteer → Playwright. Pure migration, no new features, parity confirmed first.

Then, once stable: Technical SEO / Crawling depth pack, in order:

- Duplicate content/title/meta detection across a crawl (pure aggregation over existing data)
- Hreflang validation + circular-reference detection
- Uncrawlable-link detection
- Custom data extraction (XPath/CSS/regex) — general-purpose, unlocks future checks
- Full mobile usability audit (beyond viewport tag)
- Full accessibility audit (WCAG-style) — pull in axe-core
- Spelling & grammar checks
- AMP validation
- XML sitemap generation (not just detection)
- Forms-based auth for crawling gated pages
- Visual site architecture / link graph (pairs with frontend work)

Server log file analysis is deferred to v4 — it's really an ingestion/analytics problem, not a crawler problem.

v3 — Data Layer Migration

MySQL → Postgres. Pure migration, as before.

Then: Performance & Core Web Vitals pack:

- LCP/CLS/INP/FCP/Speed Index/TBT/TTI metrics (PSI API integration)
- Weighted composite performance score
- Desktop vs. mobile split testing
- Opportunities & diagnostics fix recommendations
- Regression tracking over time — Postgres JSONB + existing scheduled-check pattern pays off here
- Lab vs. field (real-user) data comparison — needs CrUX API, saved for last in this pack

Then: the 2 remaining AI Visibility gaps:

- LLM-facing content rewrite guidance (single LLM call)
- Brand/citation tracking across AI answer engines (periodic external polling — lighter weight, good v3 fit)

v4 — Website Analytics Tool

The largest milestone in this roadmap, warranting its own sub-versioning.

v4.1 Foundation

Fastify ingestion, ClickHouse, tracking snippet, event-based visitor tracking, auto-tracked engagement events

v4.2 Visual behavior

Click/scroll/area heatmaps

v4.3 Session replay

rrweb SDK, S3 storage, replay viewer, rage-click/dead-click/thrash detection, JS error tracking tied to sessions

v4.4 Analysis tools

User journey/flow viz, funnels, path analysis, cohorts, exploration tables, custom tags/segmentation, server log file analysis

v4.5 Real-time + collaboration

Real-time dashboard, annotations, multi-site/multi-team management

v4.6 AI layer

AI session summaries + friction flagging, AI chat/natural-language querying, anomaly detection, AI-generated insights

v4.7 Advanced GA4-tier

Attribution modeling (rules-based → data-driven ML), predictive metrics, e-commerce funnels, consent-mode modeling, BigQuery/raw export

v4.7 is genuinely the deepest set of items in the whole gap list — placed last for that reason.

v5 — Production Infra + Productization

Fargate/RDS/ElastiCache/S3 cutover; Go ingestion if load demands it (as before).

Then: Platform & Integration pack:

- Public, documented API
- GDPR/CCPA input masking
- GA / Search Console / PSI integrations
- MCP server
- CI/CD integration (Lighthouse-CI style)
- BI / Looker Studio connectors

Parallel Backlog (Not Version-Gated)

Keyword/Backlink Intelligence and Content & AI Writing sit outside this roadmap's dependency chain. I'd propose scoping and building these whenever capacity allows, independent of where we are in v2–v5.

- Keyword/Backlink Intelligence (9 gaps) — requires a third-party backlink data provider; own vertical
- Content & AI Writing (8 gaps) — requires LLM API calls; own vertical

Proposed Next Step

I'd like to formalize this as a tracked plan — either as GitHub Milestones with one issue per feature grouped under each version, or as a living checklist doc — and get sign-off to begin v2 (backend framework migration) as the next unit of work.